# Linear Algebra

## Formula sheet + conditions

A formula is useful only with its assumptions. This sheet pairs each relationship with a reminder of a limitation or common mistake. Read the linked lesson notes for derivations and examples. Symbols: $A^T$ is transpose, $A^{-1}$ inverse, $A^+$ pseudoinverse, $||\cdot||_2$ Euclidean/operator 2-norm, I identity, λ eigenvalue, σ singular value. Derivatives use radians and natural logarithms unless stated otherwise.

### Vector spaces and dimensions

$\dim \text{null}(A) = n - \text{rank}(A)$ for $A \in \mathbb{R}^{m \times n}$

$\dim \text{null}(A^T) = m - \text{rank}(A)$

$\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$

A basis is an independent spanning set. Use pivot columns of the original matrix for a column-space basis.

### Determinants and inverse identities

$\det(cA) = c^n \det(A)$ for $n \times n$ $A$

$\det(AB) = \det(A)\det(B)$; $\det(A^T) = \det(A)$

$(AB)^{-1} = B^{-1}A^{-1}$

$[[a,b],[c,d]]^{-1} = [[d,-b],[-c,a]]/(ad-bc)$

The inverse formulas require invertible factors or a nonzero determinant. Row swap negates det; row scaling multiplies it; row replacement preserves it.

### Rank and consistency

$\text{rank}(AB) \le \min(\text{rank } A, \text{rank } B)$

$\text{rank}(AB) \ge \text{rank } A + \text{rank } B - n$ when $A$ has $n$ columns and $B$ has $n$ rows

$\text{rank}(A+B) \le \text{rank } A + \text{rank } B$

$Ax = b$ consistent $\Leftrightarrow \text{rank}(A) = \text{rank}([A|b])$

A consistent n-variable system has $n - \text{rank}(A)$ free variables. Full column rank gives uniqueness when a solution exists.

### Orthogonal projection and QR

$\text{proj}_a b = a(a^T b)/(a^T a)$, $a \neq 0$

$P = A(A^T A)^{-1} A^T$ for independent columns

$P = QQ^T$ for orthonormal columns

$v_j = a_j - \Sigma_{i<j}(q_i^T a_j) q_i$; $q_j = v_j / ||v_j||$

The last line is Gram-Schmidt; a zero residual signals dependence. For an orthogonal projector, $P^T = P$ and $P^2 = P$.

## Spectral and quadratic identities

$\det(A-\lambda I)=0$; $\Sigma\lambda_i=\operatorname{tr}(A)$; $\Pi\lambda_i=\det(A)$

$A=S\Lambda S^{-1} \Rightarrow A^k=S\Lambda^k S^{-1}$

$A=A^T \Rightarrow A=Q\Lambda Q^T$, $Q^TQ=I$

$\lambda_{min}||x||^2 \le x^TAx \le \lambda_{max}||x||^2$

Diagonalization needs an eigenbasis. The last two statements assume a real symmetric matrix. SPD means all eigenvalues positive; PSD means all nonnegative.

## Factorization and singular values

$PA=LU \Rightarrow Ly=Pb$, $Ux=y$

$A=U\Sigma V^T$; $\sigma_i^2=\lambda_i(A^TA)$

$||A||_2=\sigma_{max}$; $||A||_F^2=\Sigma\sigma_i^2$

$A^+=V\Sigma^+U^T$; $x_{LS}=A^+b$

$||A-A_k||_2=\sigma_{k+1}$; $||A-A_k||_F^2=\Sigma_{i>k}\sigma_i^2$

LU uses the $PA=LU$ convention here. The pseudoinverse inverts only positive singular values. Truncated SVD keeps the largest k; its error formulas apply to this best low-rank approximation.

# Lesson-by-lesson recall

## Vectors & geometric intuition

$a\,u + b\,v = (a\,u_1+b\,v_1, \ldots, a\,u_n+b\,v_n)^T$

A line not passing through the origin is an affine set, not a vector subspace.

$u^Tv = \Sigma_i u_i v_i \cdot ||u||_2 = \sqrt{(u^Tu)} \cdot \cos\theta = u^Tv/(||u||_2||v||_2)$

Orthogonal means dot product zero. Orthonormal additionally requires unit length.

$A=[T(e_1) \ldots T(e_n)] \Rightarrow Ax=\Sigma_i x_i T(e_i) \cdot (AB)x=A(Bx)$

The rightmost matrix acts first when transformations act on column vectors.

## Spaces, subspaces & bases

$W \leq V \Leftrightarrow W \neq \emptyset$ and $a\,u+b\,v \in W$ for every $u,v \in W$ and $a,b \in R$

“Degree exactly d” is not a polynomial vector space: zero is excluded and cancellation can lower degree.

$[v_1 \ldots v_k]c=0$ has only $c=0 \Leftrightarrow \{v_1,\ldots,v_k\}$ is independent

Independent rows and independent columns have different counts; for a rectangular matrix, say which independence you mean.

$x=B[x]_B \cdot [x]_C=C^{-1}B[x]_B \cdot \dim(U+W)=\dim U+\dim W-\dim(U\cap W)$

An independent set need not span the whole ambient space; specify the space for which it is a basis.

## Matrix operations & blocks

$(AB)_{ij}=\Sigma_k A_{ik}B_{kj} \cdot (m\times n)(n\times p)\to m\times p$

AB=AC does not imply B=C unless cancellation by A is justified, for example A is invertible.

$(AB)^T=B^TA^T \cdot (AB)^{-1}=B^{-1}A^{-1} \cdot A^{-1}=(1/(ad-bc))[[d,-b],[-c,a]]$

A matrix with a zero diagonal entry can still be invertible. [[0,1],[1,0]] is a counterexample.

$S=D-CA^{-1}B \cdot \det([[A,B],[C,D]])=\det(A)\det(S)$ [A invertible]

A partition is a grouping device; it does not make matrix blocks commute.

## Special matrices

$A=S+K,\ S=(A+A^T)/2,\ K=(A-A^T)/2 \cdot x^TKx=0$

A zero diagonal does not imply skew-symmetry; the paired off-diagonal entries must also be negatives.

$Q^TQ=I \Rightarrow Q^{-1}=Q^T,\ ||Qx||_2=||x||_2,\ \det Q=\pm 1$

If U is tall with orthonormal columns, $U^TU=I$ does not imply $UU^T=I$.

$P^2=P$; orthogonal projection also has $P^T=P$. $P_{line}=uu^T/(u^Tu),\ u\neq 0$

Idempotence alone does not imply orthogonal projection. [[1,1],[0,0]] is idempotent but not symmetric.

## Systems & Gaussian elimination

$Ax=b$ is consistent $\Leftrightarrow b\in \mathrm{Col}(A) \Leftrightarrow \mathrm{rank}(A)=\mathrm{rank}([A|b])$

A homogeneous system is always consistent, even when A is singular.

$R_i \leftarrow R_i - (a_{ik}/a_{kk})R_k$ [nonzero pivot $a_{kk}$]

A zero pivot at the current position does not prove singularity; a row exchange may repair it.

Consistent: $x=x_p+N\,t$, where columns of N span Null(A); number of parameters=$n-\mathrm{rank}(A)$

“Determinant zero” does not distinguish zero solutions from infinitely many solutions.

## Rank, nullity & fundamental spaces

$A\in R^{m\times n}$, rank $A=r \Rightarrow$ nullity $A=n-r$, nullity $A^T=m-r$

Use pivot columns of the original matrix for a column-space basis, not the altered columns of RREF.

$\mathrm{rank}\,A+\mathrm{rank}\,B-n \le \mathrm{rank}(AB) \le \min(\mathrm{rank}\,A,\mathrm{rank}\,B)$ · $\mathrm{rank}(A^TA)=\mathrm{rank}\,A$

$\mathrm{rank}(AB)=\mathrm{rank}(A)\mathrm{rank}(B)$ is not a valid identity.

$Ax=b$ solvable $\Leftrightarrow y^Tb=0$ for every y with $A^Ty=0$

Orthogonality to one left-null vector suffices only if it spans the entire left null space; otherwise test a basis.

## Determinants & invertibility

$\det [[a,b],[c,d]]=ad-bc$ · area scale=$|\det A|$ · $\det A\neq 0 \Leftrightarrow$ A invertible

$\det(A+B)$ is not generally $\det(A)+\det(B)$.

Row add: unchanged; row swap: $\times(-1)$; one-row scale c: $\times c$; $\det(cA)=c^n\det A$

For $n\times n$ A, multiplying every entry by 2 multiplies the determinant by $2^n$, not merely 2.

$\det(I+uv^T)=1+v^Tu$ · $\det [[A,B],[0,D]]=\det A\cdot\det D$

A determinant root identifies singularity, not the entire solution behavior of $Ax=b$.

## Orthogonality & least squares

$p=a(a^Tb)/(a^Ta)$ · $P=A(A^TA)^{-1}A^T$ when A has independent columns

The projection coefficient is $a^Tb$ only when a is already a unit vector.

$v_j=a_j-\Sigma_{i<j}(q_i^Ta_j)q_i$; $q_j=v_j/||v_j||$; $A=QR$

Subtract projections on the orthonormal vectors, not blindly on the original unnormalized columns.

$\min_x ||Ax-b||^2 \Rightarrow A^T(A\hat{x}-b)=0$; full column rank $\Rightarrow \hat{x}=(A^TA)^{-1}A^Tb$

An inconsistent system can have a unique least-squares minimizer even though it has no exact solution.

## Eigenvalues & eigenvectors

$Av=\lambda v,\ v\neq 0 \Leftrightarrow \det(A-\lambda I)=0$ and $v\in \text{Null}(A-\lambda I)\{0\}$

A repeated eigenvalue does not determine how many independent eigenvectors exist.

$A=S\Lambda S^{-1} \Leftrightarrow$ n independent eigenvectors; symmetric real $A \Rightarrow A=Q\Lambda Q^T$

A repeated eigenvalue neither guarantees nor rules out diagonalizability.

$p(A)v=p(\lambda)v \cdot A^2-\text{tr}(A)A+\det(A)I=0$ for 2×2 A

Row reduction preserves rank but does not generally preserve eigenvalues; similarity does.

## Quadratic forms & curvature

$q(x)=x^TSx,\ S=S^T \cdot \text{SPD} \Leftrightarrow$ every $\lambda_i>0 \cdot \text{PSD} \Leftrightarrow$ every $\lambda_i\geq 0$

Positive diagonal entries or a positive determinant alone do not establish positive definiteness.

$\lambda\min \leq (x^TAx)/(x^Tx) \leq \lambda\max \cdot x=Qz \Rightarrow q=\Sigma\ \lambda_i z_i^2$

For a non-symmetric matrix, apply quadratic-form claims to its symmetric part; eigenvalues of the original matrix need not govern the quotient.

$\nabla f=Ax-b \cdot A>0 \Rightarrow x^*=\text{solve}(A,b) \cdot \nabla^2 \tfrac{1}{2}||Mx-y||^2=M^TM$

A stationary point is not automatically a minimum. Check curvature and, for singular matrices, whether a solution exists.

## LU, pivoting & stable solves

$A=LU \cdot l_{ik}=u_{ik}/u_{kk} \cdot Ly=b,\ Ux=y \cdot \det(A)=\Pi u_{ii}$ (unit L)

L stores elimination multipliers with their original signs, because it reverses the row subtraction.

$PA=LU \Rightarrow Ly=Pb,\ Ux=y \cdot P^{-1}=P^T \cdot \det(A)=(-1)^s\ \Pi u_{ii}$

A zero first diagonal entry does not prove singularity; try a row interchange.

$A>0 \Rightarrow A=LL^T \cdot r=b-A\hat{x} \cdot \kappa_2(A)=\sigma\max/\sigma\min$

A small determinant is scale-dependent; use singular values or a condition number to judge sensitivity.

## SVD, pseudoinverse & low-rank models

$Av_i=\sigma_i u_i \cdot \sigma_i=\sqrt{\lambda_i(A^TA)} \cdot \text{rank}(A)=\#\{\sigma_i>0\}$

Singular values are nonnegative; they are not generally the eigenvalues or absolute diagonal entries of A.

$A^+=V\Sigma^+U^T \cdot x\text{LS}=A^+b \cdot AA^+=\text{Pcol}(A) \cdot A^+A=\text{Prow}(A)$

A pseudoinverse returns a least-squares solution; it does not make an inconsistent system exact.

$A_k=\Sigma_{i\leq k}\ \sigma_i u_i v_i^T \cdot ||A-A_k||F^2=\Sigma_{i>k}\ \sigma_i^2 \cdot \text{PCA variance}=\sigma_i^2/(n-1)$

Energy uses squared singular values, not their sum. Center data before interpreting SVD directions as PCA axes.